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14th Sep. 2026 Lecture 1 Greetings&Intro

Lec 1~8 Part 1 Logic Lec 9~12 Part2 Layout

SoC: System on Chip

Synthesis step: Make new stuff (Go forward in design)

Verification step: Check that it worked (Look backward)

14th Sep. 2026 Lecture 2 Computational Boolean Algebra

Boolean objects as data structures + operators

Shannon Expansions:

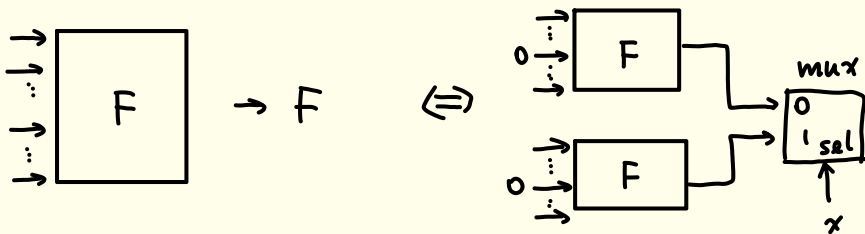
$F(x_1, x_2, \dots, x_n)$ Define x_i as constant.

$$\begin{cases} F(x_1, x_2, \dots, x_i = 0, \dots, x_n) \rightarrow F_{\bar{x}_i} \text{ (negative cofactor)} \\ F(x_1, x_2, \dots, x_i = 1, \dots, x_n) \rightarrow F_{x_i} \text{ (positive cofactor)} \end{cases}$$

Shannon Expansion Theorem:

$$F(x_1, x_2, \dots, x_n) = x_i \cdot F(x_i = 1) + x_i' \cdot F(x_i = 0) \quad (\text{Easy to prove, but powerful})$$

Another View:



Works for more than one variable:

$$\begin{aligned} F(x, y, z, w) &= x \cdot F(x=1) + \bar{x} \cdot F(x=0) \\ &= x \cdot (y \cdot F(x=1, y=1) + \bar{y} \cdot F(x=1, y=0)) + \bar{x} \cdot (y \cdot F(x=0, y=1) + \bar{y} \cdot F(x=0, y=0)) \\ &= xy \cdot F(x=1, y=1) + x\bar{y} \cdot F(x=1, y=0) + \bar{x}y \cdot F(x=0, y=1) + \bar{x}\bar{y} \cdot F(x=0, y=0) \end{aligned}$$

Remember: each cofactor is a function

AND OR XOR... Do it when the order is convenient.

$$(\bar{F}_x) = \overline{(F_x)}$$

$$(F \cdot G)_x = F_x \cdot G_x$$

$$(F + G)_x = F_x + G_x$$

$$(F \oplus G)_x = F_x \oplus G_x$$

Boolean Difference

DEFINE:

$$\frac{\partial f}{\partial x} = f_x \oplus f_{\bar{x}} \quad \text{it does not depend on } x$$

Regular Rules:

$$\textcircled{1} \frac{\partial f}{\partial x \partial y} = \frac{\partial f}{\partial y \partial x} \quad \textcircled{2} \frac{\partial (f \oplus g)}{\partial x} = \frac{\partial f}{\partial x} \oplus \frac{\partial g}{\partial x} \quad \textcircled{3} \text{ (if } f \text{ is constant) } \frac{\partial f}{\partial x} = 0$$

for "." "+" "do not" work the same, exp:

$$\begin{aligned} A + A &= A & 1+1 &= 1 \\ A \cdot A &= A \end{aligned}$$

$$\begin{aligned} \frac{\partial (f \cdot g)}{\partial x} &= (f \cdot g)_x \oplus (f \cdot g)_{\bar{x}} = (f_x \cdot g_x) \oplus (f_{\bar{x}} \cdot g_{\bar{x}}) \\ &= \left((f_{\bar{x}} \oplus \frac{\partial f}{\partial x}) \cdot (g_{\bar{x}} \oplus \frac{\partial g}{\partial x}) \right) \oplus f_{\bar{x}} g_{\bar{x}} \\ &= f_{\bar{x}} \frac{\partial g}{\partial x} \oplus g_{\bar{x}} \frac{\partial f}{\partial x} \oplus \frac{\partial f}{\partial x} \frac{\partial g}{\partial x} \\ &= f_{\bar{x}} \frac{\partial g}{\partial x} \oplus (g_{\bar{x}} \oplus \frac{\partial g}{\partial x}) \frac{\partial f}{\partial x} \\ &= f_{\bar{x}} \frac{\partial g}{\partial x} \oplus g_x \frac{\partial f}{\partial x} \end{aligned}$$

Interpreting it:

if $\frac{\partial f}{\partial x} = 1$, then the change of input x leads to the change of output.

Example:

$$\begin{aligned} a &\rightarrow \boxed{\text{1 bit}} \rightarrow s & s &= a \oplus b \oplus C_{in} \\ b &\rightarrow \boxed{\text{Full Add}} \rightarrow C_{out} & C_{out} &= ab + (a+b) C_{in} \\ C_{in} &\rightarrow \end{aligned}$$

$$C_{out}_{C_{in}} = ab + a + b \quad C_{out}_{\bar{C}_{in}} = ab$$

$$\frac{\partial C_{out}}{\partial C_{in}} = (a+b) \oplus ab = ab(\overline{a+b}) + \overline{ab}(a+b) = a \oplus b$$

Existential Quantification & Universal Quantification (Sounds like philosophy...)

Universal:

Existential:

$$\forall x_i F := F_{x_i} \cdot F_{\bar{x}_i} \quad \exists x_i F := F_{x_i} + F_{\bar{x}_i}$$

$$\downarrow$$

$$\text{if } \forall x_i F == 1$$

$$\downarrow$$

$$\text{if } \exists x_i F == 1$$

other inputs make $F=1$ There exists a value of x making $F=1$

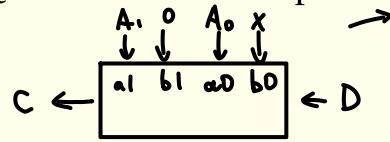
Additional Prop.

Works for more var.

$$\text{Exp. } F(x, y, z, w) \quad (\forall xy F)[z, w] = (\forall x (\forall y F)) = F_{xy} \cdot F_{\bar{x}y} \cdot F_{x\bar{y}} \cdot F_{\bar{x}\bar{y}}$$

$$(\exists xy F)[z, w] = F_{xy} + F_{\bar{x}y} + F_{x\bar{y}} + F_{\bar{x}\bar{y}}$$

Quantification Example



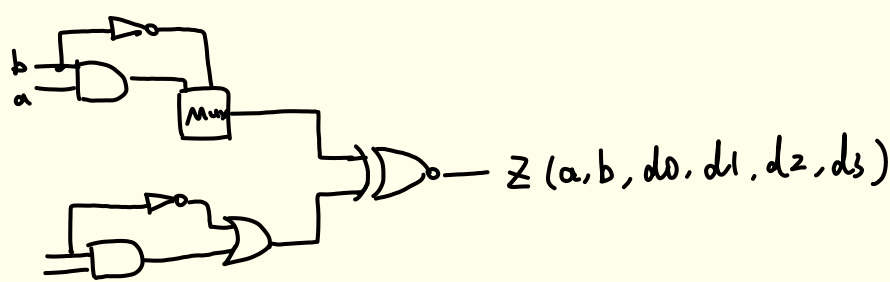
$C = A_1 A_0 X + A_1 (A_0 + X) D$

$C_{A_1 A_0} = X + D \quad C_{\bar{A}_1 A_0} = 0 \quad C_{A_1 \bar{A}_0} = X D \quad C_{\bar{A}_1 \bar{A}_0} = 0$

$\Rightarrow (\forall A_1, A_0 \ C) [X, D] = 0 \quad (\exists A_1, A_0 \ C) [X, D] = X + D + X D = X + D$ *Makes sense!*

Applied to Network repair

$f(a, b) = a \bar{b} + b$

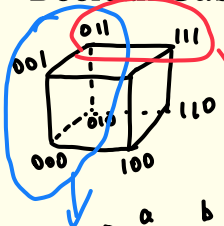


$\Rightarrow (\forall a, b \ Z) [d_0, d_1, d_2, d_3] = 1 \quad d_0 = 0, d_1 = 1, d_2 = 1, d_3 = X$

Recursive Tautology (递归重言式/递归恒真式)

Represent a function as a set of OR-ed product terms

Boolean Cube



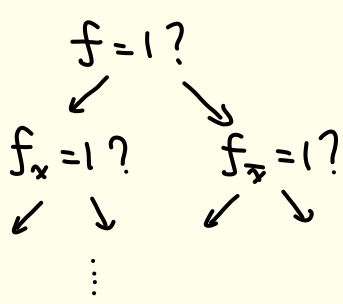
PCN: Positional Cube Notation

$X: 01 \quad \bar{X}: 10 \quad \text{no } X \text{ or } \bar{X}: 11$

$\bar{a} = [10 \ 11 \ 11] \quad bc = [11 \ 01 \ 01]$ *present AND*

Ex. $f(a, b, c) = a + bc + ab \Rightarrow [01 \ 11 \ 11], [11 \ 01 \ 01], [01 \ 01 \ 11]$ *present OR*

$f_1 = 1 \Leftrightarrow \text{both cofactors} = 1$



URP Unate Recursive Paradigm

① Mechanics

positive cofactor for x

$[...10...]$ $\rightarrow \bar{x} \rightarrow 0$ (just remove) $[...01...]$ $\rightarrow x \rightarrow 1$ make it 11 $[11]$ $\rightarrow$ leave this alone

negative cofactor for x

$[...10...]$ $\rightarrow x \rightarrow 1$ (make it 11) $[...01...]$ $\rightarrow x \rightarrow 0$ (remove) $[11]$ $\rightarrow [11]$

② Selection & termination

Unate functions (sum of product)

f is unate := its SOP only has each literal in exactly one polarity

Ex: $f(a,b,c,d,e) = ab + a\bar{c}d + \bar{c}d\bar{e}$ is unate

f is positive unate: $x: 0 \rightarrow 1 \Rightarrow$ keeps f constant / $f: 0 \rightarrow 1$

$\sim$ negative unate: $x: 0 \rightarrow 1 \Rightarrow f$ constant / $f: 1 \rightarrow 0$

f is either unate or binate

Unate cube-list for f is tautology iff it contains all don't care cube = $[11 \dots 11]$ (Rule 0)

$\dots$
11 11 $\dots$ 11 $\Rightarrow$ Tautology
 $\dots$

Rule 1: ==1 if cube-list has all don't care cube

Rule 2: !=1 if cube-list unate and all don't care cube missing

Rule 3: ==1 if cube-list has single var cube that appears in both polarities $(x + \bar{x} = 1)$